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Introduction

- Road accidents are a major cause of injuries and fatalities.
- Immediate medical assistance can save lives.
- This project automatically detects accidents and sends alerts to emergency contacts.
- It uses IoT technology for faster response and improved safety.

Empathy

- Accident victims may be unconscious or unable to seek help.
- Delays in emergency response can worsen injuries.
- Family members often remain unaware of the accident location.
- The system is designed to provide immediate assistance by sending alerts automatically. paragraph text

Problems identified

- Many road accident victims are unable to call for help due to severe injuries or unconsciousness.
- Delay in informing emergency services leads to loss of valuable rescue time.
- Accidents occurring in remote or isolated areas may go unnoticed for a long time.
- Family members and emergency contacts often do not know the exact accident location.
- Dependence on eyewitnesses to report accidents can delay medical assistance.
- Lack of an automatic accident detection and notification system increases the risk of fatalities.
- Traffic congestion and communication delays can further slow emergency response.



Ideation and Problem definition

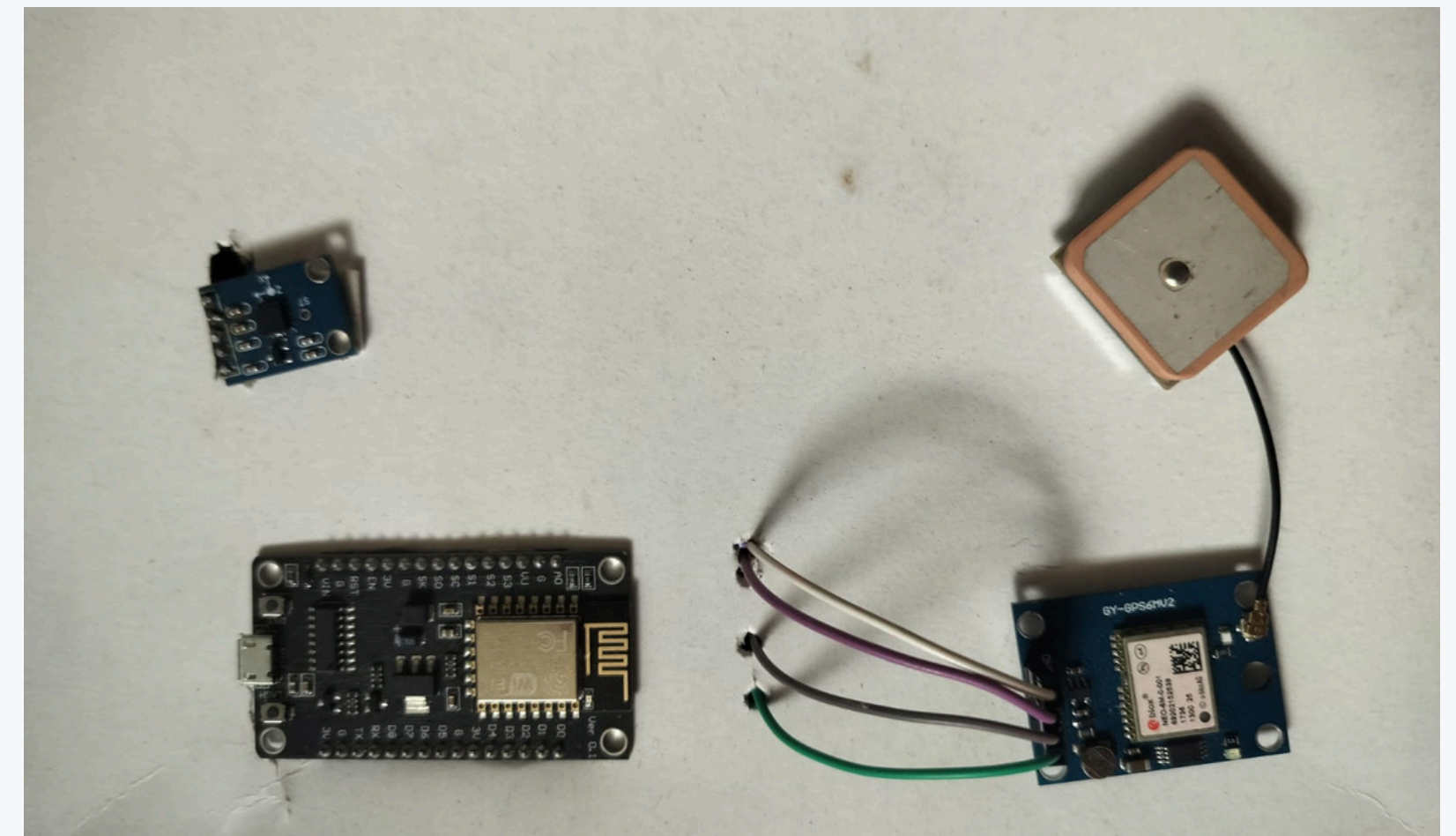
- Develop a Smart Accident Detection and Alert System using MPU6050, GPS, and GSM modules. The system automatically detects accidents and sends location-based emergency alerts to contacts.
- Road accident victims may be unable to seek help, causing delays in emergency response. An automated system is needed to detect accidents and instantly share the accident location.

Components used

- NodeMCU
- Analogue Accelerometer (GY61)
- NEO-6M gps

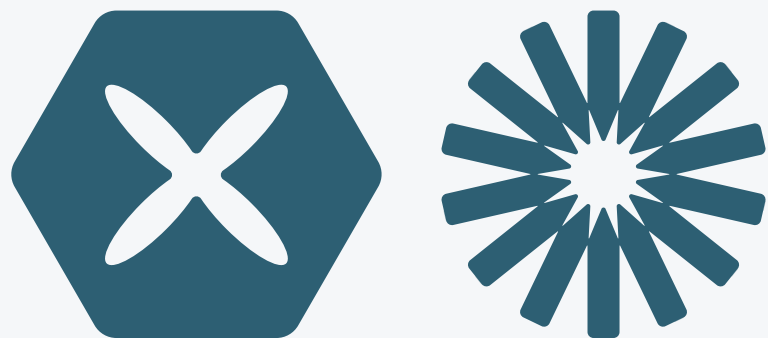
Prototype

- The NodeMCU (ESP8266) acts as the main controller and coordinates the operation of all connected module
- The GY-61 (ADXL335) accelerometer continuously monitors vehicle acceleration along the X, Y, and Z axes.
- The sensor data is continuously analyzed by the NodeMCU to detect sudden impacts or abnormal movements.
- When the acceleration exceeds a predefined threshold, the system identifies it as a possible accident.
- The Neo-6M GPS module is activated to obtain the current location of the vehicle in terms of latitude and longitude.
- The GPS antenna receives satellite signals and helps the GPS module determine the vehicle's position accurately.
- The NodeMCU processes the accident information and location data to generate an emergency alert.
- The alert containing the accident status and location details can be sent to emergency contacts, enabling a faster response and improving road safety.



Future Scope

- **Integrate the system with nearby hospitals and ambulance services for immediate medical assistance.**
- **Automatically identify and notify the nearest hospital based on the accident location.**
- **Develop a mobile application for real-time accident monitoring and tracking.**
- **Implement cloud-based storage for accident records and analysis.**
- **Enhance the system with AI-based accident detection for improved accuracy.**



Conclusions

The Smart Accident Detection and Alert System is an effective solution for improving road safety. It automatically detects accidents, obtains the accident location using GPS, and sends alert messages to family members. The system helps reduce emergency response time and can contribute to saving lives through timely assistance.



REFERENCE:

1] Dr. C. K. Gomathy¹, K Rohan², Bandi Mani Kiran Reddy³, Dr. V Geetha, "ACCIDENT DETECTION AND ALERT SYSTEM", JOURNAL OF ENGINEERING, COMPUTING & ARCHITECTURE.

2] Nicky Kattukkaran, Arun George, Mithun Haridas T.P, "Intelligent Accident Detection and Alert System for Emergency Medical Assistance", 2017 International Conference on Computer Communication and Informatics (ICCCI-2017), Jan. 05-07, 2017, Coimbatore, INDIA.



"Technology cannot prevent every accident, but it can ensure that help arrives when it is needed most."



Thank You

